

What is a Fukaya category?

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- Geometry and physics background: Symplectic geometry, Lagrangian boundary conditions
- Foundations: Floer theory, higher compositions, curvature, wrapping
- Computational techniques: Split-generation, sheaves, mirror symmetry

A “What is X?” seminar talk

Symplectic manifolds

A **symplectic manifold** is a pair (M^{2n}, ω) , where ω is a differential two-form satisfying

$$d\omega = 0, \quad \omega^n \text{ is nowhere zero.}$$

A Hamiltonian H determines vector field X_H by $\iota_{X_H}\omega = dH$.

Classical mechanics

A configuration space Q has phase space T^*Q .

$$\omega = \sum_i dq_i \wedge dp_i.$$

Hamilton, 1834–35. Kähler, 1933.

Complex geometry

A Kähler manifold has a symplectic form compatible with its complex structure. For example, a complex submanifold $X \subset \mathbb{CP}^N$ carries

$$\omega = \omega_{\text{FS}}|_X.$$

The spherical pendulum and reduction

Particle const'd to a sphere by a rigid rod

$$M = T^*S^2, \quad H(q, p) = \frac{1}{2}|p|^2 + gq_3.$$

Rotation around the vertical axis gives the conserved angular momentum

$$J(q, p) = (q \times p)_3, \quad \{H, J\} = 0.$$

The regular compact connected joint levels are Lagrangian tori.



Meyer, 1973. Marsden–Weinstein, 1974. Unit-sphere normalization: $\int_{\mathbb{CP}^1} \omega_{\text{FS}} = \pi$.

Symplectic reduction

For a regular free moment level of a Hamiltonian G -action,

$$M//G = \mu^{-1}(0)/G.$$

The Fubini–Study form

Reduce $(\mathbb{C}^{n+1}, \omega_0 = \sum_j dx_j \wedge dy_j)$ by scalar circle rotation at $|z|^2 = 1$. The quotient is $\pi : S^{2n+1} \rightarrow \mathbb{CP}^n$.

$$\pi^* \omega_{\text{FS}} = \omega_0|_{S^{2n+1}}.$$

Coisotropics in an energy level

A submanifold $C \subset M$ is called a **coisotropic** if

$$(T_x C)^\omega \subset T_x C \quad (x \in C).$$

Preservation by Hamiltonian flow

If $H|_C$ is constant, the Hamiltonian flow preserves C .

Proof. For $v \in T_x C$,

$$\omega(X_H, v) = dH(v) = 0 \implies X_H(x) \in (T_x C)^\omega \subset T_x C.$$

In fact, trajectories stay in the leaves of the characteristic distribution $(TC)^\omega$.

Lagrangian submanifolds

A half-dimensional submanifold $L^n \subset (M^{2n}, \omega)$ is **Lagrangian** if

$$\omega|_L = 0, \quad \text{equivalently} \quad (TL)^\omega = TL.$$

The Lagrangian case

If L is closed as a subset and lies in one energy level of H , then the Hamiltonian flow preserves L .

Hamilton–Jacobi theory seeks a Lagrangian foliation whose leaves lie in energy levels of H .

- The zero section and each cotangent fiber in T^*Q .
- The graph of a closed one-form on Q .
- Regular torus fibers of a completely integrable system (e.g. the spherical pendulum).
- Smooth curves in a symplectic surface.

Open strings and the A-model

An open string stretching from the Lagrangian L_0 to L_1 is a path $\gamma : [0, 1] \rightarrow M$ with

$$\gamma(0) \in L_0, \quad \gamma(1) \in L_1.$$

- A worldsheet is a map $u : \Sigma \rightarrow M$, where Σ is a compact Riemann surface with marked points on its boundary. Each boundary segment maps to a Lagrangian.
- The topological A-twist of the supersymmetric sigma model extracts a theory governed by symplectic geometry.
- Its amplitudes localize to pseudoholomorphic curves.

Boundary conditions become objects. Open-string states become morphisms. Joining strings gives compositions.

Witten, topological sigma models, 1988.

Witten's supersymmetric quantum mechanics

For a particle in $\mathbb{R}^n$, choose a function f and set $V = \frac{1}{2}|\nabla f|^2$. The real-time Lagrangian of the flat target model is

$$L_{\text{SUSY}} = \underbrace{\frac{1}{2}|\dot{x}|^2 - \frac{1}{2}|\nabla f|^2}_{L_{\text{ordinary}}} + \underbrace{\frac{i}{2}(\bar{\psi}_i \dot{\psi}^i - \dot{\bar{\psi}}_i \psi^i) - f_{ij} \bar{\psi}^i \psi^j}_{\text{fermionic terms}}.$$

Here $f_{ij} = \partial_i \partial_j f$ and $\psi, \bar{\psi}$ are anticommuting variables.

Ordinary quantum mechanics

Omitting the fermionic variables gives a particle in the potential V . Quantization, with mass and $\hbar$ equal to 1, gives

$$H_{\text{ordinary}} = -\frac{1}{2}\Delta + V \quad \text{on scalar wavefunctions.}$$

Witten, 1982. On curved targets, the supersymmetric Lagrangian includes connection and curvature terms.

Supersymmetry and Morse theory

For a configuration manifold Q , quantizing the fermions gives differential forms:

$$\mathcal{H}_{\text{ordinary}} = L^2(Q), \quad \mathcal{H}_{\text{SUSY}} = L^2\Omega^*(Q).$$

The supersymmetric Hamiltonian includes Hessian terms acting on differential forms.

- On a closed Riemannian manifold, the zero-energy ground states represent $H^*(Q)$.
- For Morse f and large deformation parameter, critical points index a basis of the low-energy complex.
- Tunneling along gradient trajectories gives the Morse differential.

Fukaya's extension

For several functions, gradient trees for their differences define higher compositions between Morse complexes. These operations satisfy Stasheff's A_∞ relations.

Witten, 1982. Fukaya, 1993. Fukaya–Oh, 1997.

Lagrangian Floer complex

For transverse L_0, L_1 , intersection points are the “ground states before tunneling”:

$$CF^*(L_0, L_1) = \bigoplus_{x \in L_0 \cap L_1} \Lambda \langle x \rangle.$$

The differential counts rigid strips $u : \mathbb{R} \times [0, 1] \rightarrow M$:

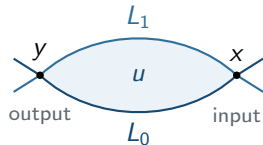
$$\begin{aligned} \partial_s u + J(u) \partial_t u &= 0, \\ u(s, 0) &\in L_0, \quad u(s, 1) \in L_1. \end{aligned}$$

Here J is an auxiliary choice of an ω -compatible almost complex structure.

- The strip converges to y as $s \rightarrow +\infty$ and to x as $s \rightarrow -\infty$.
- Broken strips give $d^2 = 0$ when disc bubbling is absent or corrected.
- Small graphs in T^*Q recover Morse theory.

Gromov, 1985. Floer, 1988.

Fukaya categories



$$\mu^1(x) = \dots \pm T^{\omega(u)} y + \dots$$

The Fukaya category: objects and coefficients

A Lagrangian brane

- A Lagrangian $L \subset M$
- A grading, when available
- Orientation / relative spin data
- A local system
- Unobstructedness data

Novikov coefficients

$$\Lambda = \left\{ \sum_i a_i T^{\lambda_i} : \lambda_i \rightarrow +\infty \right\}.$$

Here $a_i \in \mathbb{k}$, $\lambda_i \in \mathbb{R}$. A curve contributes

$$\pm T^{\int u^* \omega}$$

times a holonomy factor for the local-systems.
Gromov compactness makes infinite sums meaningful.

Morphisms are Floer cochains with degrees, signs, and local-system coefficients determined by brane data.

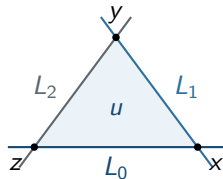
Fukaya, 1993. We use characteristic-zero Novikov coefficients for the compact examples.

Higher compositions

The category carries operations of degree $2 - k$:

$$\mu^k : CF^*(L_{k-1}, L_k) \otimes \cdots \otimes CF^*(L_0, L_1) \longrightarrow CF^*(L_0, L_k)[2 - k].$$

- μ^1 counts strips and defines the differential.
- μ^2 counts triangles and gives composition.
- μ^3 gives a homotopy between the two triple compositions.
- All μ^k together satisfy the A_∞ relations.



$$\mu^2(y, x) = \cdots \pm T^{\omega(u)} z + \cdots$$

The relations come from boundaries of one-dimensional moduli spaces of holomorphic polygons.

On HF^* , μ^2 is associative. Higher operations encode additional information, e.g. Massey prod.

Stasheff, 1963. Fukaya, 1993. Fukaya–Oh, 1997.

Homological mirror symmetry

A-side

A symplectic manifold (X, ω)

Lagrangian branes and Floer theory

B-side

An algebraic mirror $X^\vee$

Coherent sheaves and derived morphisms (Ext groups)

$$D^\pi \mathcal{F}(X, \omega) \simeq D^b \text{Coh}(X^\vee)$$

This is an exact equivalence of triangulated categories, which can usually be enhanced to a chain level equivalence.

Kontsevich, ICM 1994 (proceedings 1995). SYZ, 1996: the torus-fibration picture.

Curvature

For a graded Lagrangian L , counting pseudoholomorphic discs with boundary on L defines an element

$$\mu_L^0 \in \mathrm{CF}^2(L, L),$$

called the **curvature**.

- A strip can degenerate by bubbling off such a disc.
- Consequently $(\mu^1)^2 = 0$ is no longer automatic.
- The extra boundary strata modify the A_∞ relations, giving a **curved A_∞ -category**.

Historical correction

Kontsevich's printed ICM article includes a note added in proof introducing m_0 and correcting the uncurved description.

Oh, 1993 and 1995. Kontsevich, 1995, p. 138. Further history in the appendix.

Bounding cochains

A degree-one element b of positive valuation solves the Maurer–Cartan equation if

$$\mu_L^{0;b} = \mu_L^0 + \mu^1(b) + \mu^2(b, b) + \mu^3(b, b, b) + \cdots = 0.$$

- Insert b at all possible boundary inputs to deform the operations.
- The pair (L, b) is then an unobstructed object. These form an uncurved A_∞ -category.
- Not every Lagrangian brane admits a bounding cochain and for a given Lagrangian brane different bounding cochains often give non-isomorphic objects.
- Weak unobstructedness allows $\mu_L^{0;b} = W_L(b)e_L$. Objects with the same potential value form an A_∞ -category.

Fukaya–Oh–Ohta–Ono: preliminary manuscript 2000, monograph 2009.

Virtual techniques

The moduli spaces required for Floer theory need not be smooth of the expected dimension.

- Multiply covered curves obstruct naive transversality.
- Virtual perturbation methods define virtual counts and chains compatible with gluing.
- Compactification boundaries still determine the algebraic identities.

Two separate issues

Virtual methods construct the curve counts. Bounding cochains solve the curvature equation for an object.

Liu–Tian, 1998. Fukaya–Ono, 1999. FOOO, 2000/2009.

Split-generation

Take the smallest completion of the category of geometric branes that admits
shifts, finite sums, cones, and direct summands.

Definition

Branes $L_1, \dots, L_r$ **split-generate** if every object lies in their closure under these operations.

Set $G = \bigoplus_i L_i$ and retain the full A_∞ endomorphism algebra

$$A = \mathrm{CF}^*(G, G).$$

$D^\pi \mathcal{F}(M) \simeq H^0 \mathrm{Perf}(A)$. A finite collection of Lagrangians can determine the entire category.

Seidel, 2008. Abouzaid's generation criterion, 2010.

Structural properties and generation criteria

- Hamiltonian isotopy preserves objects up to quasi-isomorphism.
- For compact branes, Floer cohomology is finite-dimensional.
- In an oriented graded $2n$ -dimensional setting, Floer duality gives

$$\mathrm{HF}^k(L_0, L_1) \cong \mathrm{HF}^{n-k}(L_1, L_0)^\vee.$$

- Lagrangian surgery and Dehn twists produce exact triangles.

Open–closed generation criterion

For a full subcategory $\mathcal{A}$, consider

$$\mathrm{OC} : \mathrm{HH}_{*-n}(\mathcal{A}) \longrightarrow \mathrm{QH}^*(M), \quad 1 \in \mathrm{im}(\mathrm{OC}),$$

Under the standard hypotheses, hitting the unit implies split-generation.

Abouzaid, 2010. Ganatra, 2013 and 2016.

The wrapped Fukaya category

Let M be a Liouville manifold and allow exact Lagrangians conical at infinity.

$$\mathrm{HW}^*(L_0, L_1) = \varinjlim_{w \rightarrow \infty} \mathrm{HF}^*(L_0, L_1; H_w).$$

- The Hamiltonians wrap the ends of the Lagrangians.
- Morphisms detect Hamiltonian chords far out at infinity.
- Polygon counts define the wrapped Fukaya category $\mathcal{W}(M)$.
- One may optionally add stops to restrict which wrapping is allowed.

For a Weinstein manifold, cocores of the critical handles provide split-generators.

Abouzaid–Seidel, 2007 preprint / 2010. Cocore generation: Chantraine et al.; Ganatra–Pardon–Shende.

Cotangent bundles and based loops

For Q closed, connected, and spin, a cotangent fiber $F_q = T_q^*Q$ split-generates $\mathcal{W}(T^*Q)$.

$$\mathrm{CW}^*(F_q, F_q) \simeq C_{-*}(\Omega_q Q).$$

The product corresponds to concatenation of based loops.

The circle

For $Q = S^1$, the components of $\Omega_q S^1$ are indexed by $\mathbb{Z}$ and contractible. Hence

$$\mathrm{HW}^*(F_q, F_q) \cong \mathbb{k}[z, z^{-1}], \quad D^\pi \mathcal{W}(T^*S^1) \simeq \mathrm{Perf}(\mathbb{G}_m).$$

The wrapped Floer cohomology is concentrated in degree zero.

Abouzaid, 2009 preprint / 2012 (based loops), 2010 preprint / 2011 (generation).

Infinitesimal wrapping and constructible sheaves

Nadler–Zaslow and Nadler

For a compact real analytic manifold Q , the suitable category of infinitesimally wrapped branes satisfies

$$D^\pi \mathcal{F}_{\text{inf}}(T^*Q) \simeq D_c^b(Q).$$

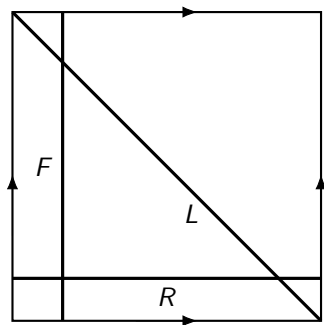
Here $D_c^b(Q)$ is the bounded derived category of constructible sheaves: cohomology sheaves are locally constant of finite rank on each stratum.

- A zero section with local system corresponds to that local system.
- A cotangent fiber corresponds to a skyscraper sheaf.
- Asymptotic Lagrangian directions correspond to microsupport.

There is also a theory of wrapped sheaves that matches the wrapped Fukaya category.

Nadler–Zaslow and Nadler, 2006 preprints / 2009. Ganatra–Pardon–Shende, 2018 / 2024.

The symplectic two-torus



Any two split-generate $\mathcal{F}(T^2, \omega)$.

Put $A = \int_{T^2} \omega$ and $q = T^A$.

The mirror elliptic curve

$$E_q = \{y^2z + xyz = x^3 + a_4(q)xz^2 + a_6(q)z^3\} \subset \mathbb{P}_\Lambda^2,$$

$$E_q^{\text{an}} = \mathbb{G}_m^{\text{an}} / q^{\mathbb{Z}}.$$

Under HMS:

- $R \longleftrightarrow \mathcal{O}_{E_q}$
- $F \longleftrightarrow \mathcal{O}_p$
- An “exact” surgery circle L corresponds to $\mathcal{O}_{E_q}(p)$.

Polishchuk–Zaslow, 1998. Lekili–Perutz, 2012 (arithmetic refinement).

Triangles and theta functions

Lift the circles to straight lines in $\mathbb{R}^2$. For trivial local systems, a triangle product takes the form

$$\mu^2(y, x) = \left(\sum_{n \in \mathbb{Z}} \epsilon_n T^{\frac{A}{2}(n+a)^2} \right) z, \quad \epsilon_n \in \{\pm 1\}.$$

Here a is the normalized relative displacement of the lifted circles ($a \in \mathbb{Z}$ means they pass through the same point).

- Spin structures determine the signs. Nontrivial local systems add holonomy factors.
- The series is a theta function after a change of local frame.
- The triangle product matches composition of morphisms on E_q .

$$D^\pi \mathcal{F}(T^2, \omega) \simeq D^b \text{Coh}(E_q)$$

Polishchuk–Zaslow, 1998.

A quartic K3 surface

$$X_4 = \{z_0^4 + z_1^4 + z_2^4 + z_3^4 = 0\} \subset \mathbb{CP}^3, \quad \omega = \omega_{\text{FS}}|_{X_4}.$$

A smooth quartic has complex dimension 2 and trivial canonical bundle.

Homological mirror symmetry

$$D^\pi \mathcal{F}(X_4, \omega) \simeq D^b \text{Coh}(Y_4).$$

Here Y_4 is the mirror K3, obtained from the quartic pencil by the appropriate finite quotient and resolution.

The deformed A_∞ algebra of vanishing cycles is matched with a full subcategory on the mirror. Generation upgrades this comparison to an equivalence.

Seidel, 2003 preprint / 2015.

Three Lagrangian spheres

The Fukaya category of a quartic K3
is split-generated by **three Lagrangian spheres**.

For suitable branes $S_0, S_1, S_2 \cong S^2$,

$$D^\pi \mathcal{F}(X_4) = \langle S_0, S_1, S_2 \rangle_{\text{split}}.$$

- A very ample line bundle $\mathcal{A}$ on the projective mirror gives split-generators $\mathcal{O}, \mathcal{A}, \mathcal{A}^2$.
- In the K3 HMS construction, these line bundles have Lagrangian sphere representatives.

Orlov, 2009, Theorem 4. Sheridan–Smith, 2017/21 (HMS).

Hacking–Keating, 2025, §1.4 and Corollary 2.50.

The quintic threefold

$$X_5 = \{z_0^5 + z_1^5 + z_2^5 + z_3^5 + z_4^5 = 0\} \subset \mathbb{CP}^4.$$

This is a Calabi–Yau threefold with the restricted Fubini–Study form.

Homological mirror symmetry

$$D^\pi \mathcal{F}(X_5, \omega) \simeq D^b \text{Coh}(Y_5).$$

The original curve-counting prediction (1991)

Candelas–de la Ossa–Green–Parkes used periods of the mirror to predict the quintic’s genus-zero Gromov–Witten theory.

$$n_1 = 2875, \quad n_2 = 609250, \quad n_3 = 317206375.$$

The instanton numbers n_d match naive rational-curve counts on a general quintic for $d < 10$.

Candelas et al., 1991. Nohara–Ueda, 2011/12. Sheridan, 2011/15.

How HMS recovers the curve counts

The genus-zero Yukawa coupling of the quintic is

$$K(q) = 5 + \sum_{d \geq 1} n_d d^3 \frac{q^d}{1 - q^d}.$$

The 5 is the classical triple intersection number. The denominator records multiple covers.

- Enhanced HMS identifies the categories' cyclic homology, with its parameter dependence.
- Cyclic open–closed gives the quantum connection on the A-side. The B-side gives the Gauss–Manin connection on de Rham cohomology.
- Matching the connections and pairings identifies the Yukawa couplings, in the flat mirror coordinate q with normalized volume form.

A suitable enhancement of HMS for the quintic implies the 1991 predictions.

Ganatra–Perutz–Sheridan, 2015. Ganatra–Sheridan, 2025.

The Fukaya category

- Its objects are Lagrangian boundary conditions with brane and unobstructedness data.
- Its morphisms and higher compositions come from holomorphic curves.
- Curvature records disc bubbling. Bounding cochains correct it when possible.
- A finite set of split-generators determines the formal completion $D^\pi \mathcal{F}(M)$ through its A_∞ endomorphism algebra.
- Homological mirror symmetry identifies this completion with a derived category of coherent sheaves on a mirror.
- For noncompact manifolds, wrapped and infinitesimally wrapped categories include Lagrangians extending to infinity.

Appendix: the history of curvature

- **Floer, 1988:** hypotheses excluding disc bubbling, such as $\pi_2(M, L) = 0$.
- **Oh, 1993:** monotone Floer theory with minimal Maslov number at least 3 excludes disc bubbles in the relevant low-dimensional moduli spaces.
- **Kontsevich, 1995:** the printed ICM article explicitly introduces m_0 in its note added in proof.
- **Oh, 1995:** the Maslov-two addendum establishes the cancellation needed for self-Floer theory.

Disc bubbling was already an analytic issue. Curvature gives its algebraic expression inside the Fukaya category.

Floer, JDG 28 (1988). Oh, CPAM 46 (1993), 48 (1995). Kontsevich, ICM proceedings (1995).

Kontsevich's note added in proof

The ICM address took place in 1994. The proceedings appeared in 1995.

- The printed article, p. 138, corrects the omission of a disc-bubbling boundary stratum.
- It introduces degree-two elements m_0 and modifies the A_∞ identities.
- The note cites Oh's work.
- This note is absent from the November 1994 arXiv version.

The note recognizes the curved structure. The systematic obstruction and deformation theory was developed subsequently.

Kontsevich, *Homological algebra of mirror symmetry*, ICM 1994, vol. I, pp. 120–139.

Nonzero curvature and self-Floer cohomology

Suppose two weakly unobstructed objects have scalar curvature

$$\mu_{L_i}^{0;b_i} = W_i e_i.$$

With compatible sign conventions, their deformed differential satisfies

$$(d^{b_0,b_1})^2 = (W_1 - W_0) \text{id}.$$

- Equal potential values give a genuine cochain complex.
- For self-Floer theory, the two bubbling contributions cancel.
- Thus $d^2 = 0$ does not imply $\mu^0 = 0$.

Oh's Maslov-two addendum, 1995. Oh–Fukaya, 2006, §5.2.

Bounding cochains: the chronology

- **July 1996:** Fukaya, Oh, Ohta, and Ono begin their collaboration at Ascona.
- **1997:** their account credits a suggestion of Kontsevich concerning obstruction classes and bounding-chain deformation.
- **1998:** writing of the book begins.
- **December 2000:** preliminary manuscript.
- **2009:** *Lagrangian Intersection Floer Theory: Anomaly and Obstruction* appears.

$$\mu^0 + \mu^1(b) + \mu^2(b, b) + \cdots = 0.$$

FOOO, Preface and §1.6. Oh–Fukaya, 2006, §3.4 and reference [K2].

Unobstructedness and scalar curvature

- **Exactness.** If $\omega = d\lambda$ and $\lambda|_L = df$, a disc has

$$\int_D u^* \omega = \int_{\partial D} u^* \lambda = 0.$$

Hence there are no nonconstant compatible holomorphic discs.

- **Tautological unobstructedness.** For a suitable compatible J , L bounds no nonconstant J -holomorphic discs. An essential embedded circle in T^2 is an example.
- **A bounding cochain.** Discs may exist, but b solves $\mu^{0;b} = 0$.
- **Weak unobstructedness.** Scalar curvature remains, while equal-potential morphisms have $d^2 = 0$.

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Dates of the form 2003/06 mean preprint / publication.

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